

Continuous-Time Bandpass Filter Design, Modeling & Analysis Tool on MATLAB

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Abstract—This paper presents a comprehensive MATLAB-based Graphical User Interface (GUI) tool designed to automate the synthesis, analysis, and simulation of high-order Continuous-Time (CT) bandpass filters. Although discrete-time filters offer high precision for ADC applications, CT filters remain paramount for high-frequency applications such as RF front-ends. However, designing CT filters is highly iterative due to physical non-idealities like finite open-loop gain, which leads to bandwidth widening and Q-factor degradation. The proposed tool bridges the gap between ideal mathematical synthesis and physical implementation. It automatically synthesizes Cascaded Resonator Feed-Forward (CRFF) topologies for 2nd, 4th, and 6th-order filters while incorporating finite gain models. Furthermore, it accelerates the verification process by generating fully wired Simulink models for Transient & AC responses, with the option to export a current-mode Verilog-A behavioral code for easy implementation into various design suites such as Cadence Virtuoso. Simulation results validate the tool’s capability to accurately predict Q-reduction and ensure system stability before a circuit-level implementation.

Index Terms—Continuous-Time Filters, Bandpass Filter, Cascaded Resonator Feed-Forward, MATLAB, Simulink, GUI, Verilog-A, Quality Factor, Cadence, Virtuoso.

I. INTRODUCTION

Continuous-Time (CT) filters are fundamental building blocks in modern analog and mixed-signal systems, particularly in applications where sampling rates for discrete-time (switched-capacitor) or digital filters would be prohibitively high, such as RF transceivers, anti-aliasing stages, and high-speed data converters [1].

Despite having speed advantages, the design of CT filters poses significant challenges. They are highly sensitive to component variations and the inherent non-idealities of active devices. A primary limitation arises from the Operational Amplifiers used to construct the filter’s core integrators. Real OPAMPs exhibit finite open-loop DC gain, which introduces a “leakage” effect in the integration phase [2]. In high-Q bandpass filters, this finite gain causes bandwidth widening and hence, a drastic reduction in the effective Quality Factor.

Traditionally, designers rely on ideal mathematical models [3] during the initial synthesis phase, only to discover severe performance deviations during transistor-level SPICE simulations. This gap forces an even more tedious and iterative re-design loop in CT filter applications, given that the DT design methodology is better established and applied throughout the years.

To address these challenges, this paper introduces the **Continuous-Time Bandpass Filter Design & Analysis Tool**.

Developed in MATLAB, the tool provides a complete workflow shown in Fig. 1.

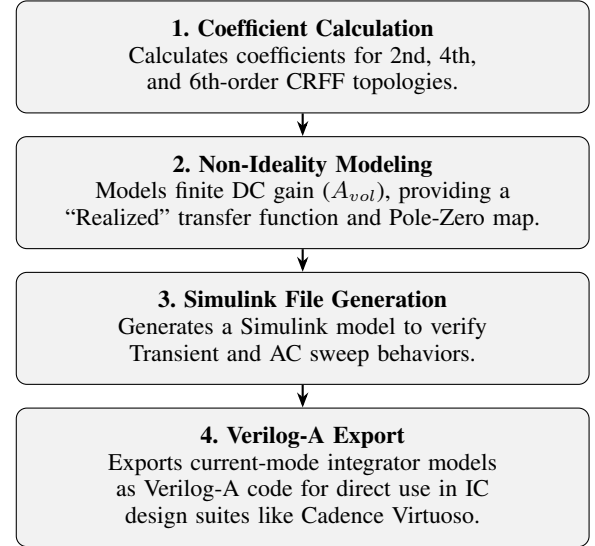


Fig. 1. Automated design and analysis workflow of the proposed tool.

The remainder of this paper is organized as follows. Section II outlines the software implementation and model generation algorithms. Section III details the theoretical framework, including the CRFF topology and integrator modeling. Section IV presents the simulation results, and Section V concludes the paper.

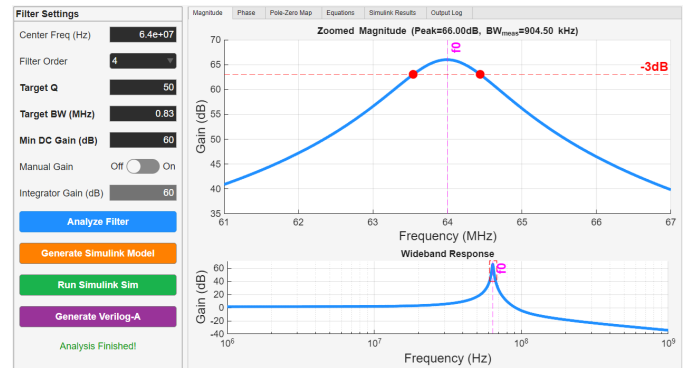


Fig. 3. The main GUI displaying the Analytical Magnitude Response. The interface features a parameter grid and visualization pane, with the top displaying the passband and measured -3dB bandwidth, while the bottom exhibits the wideband response. The Phase Response is on the second tab.

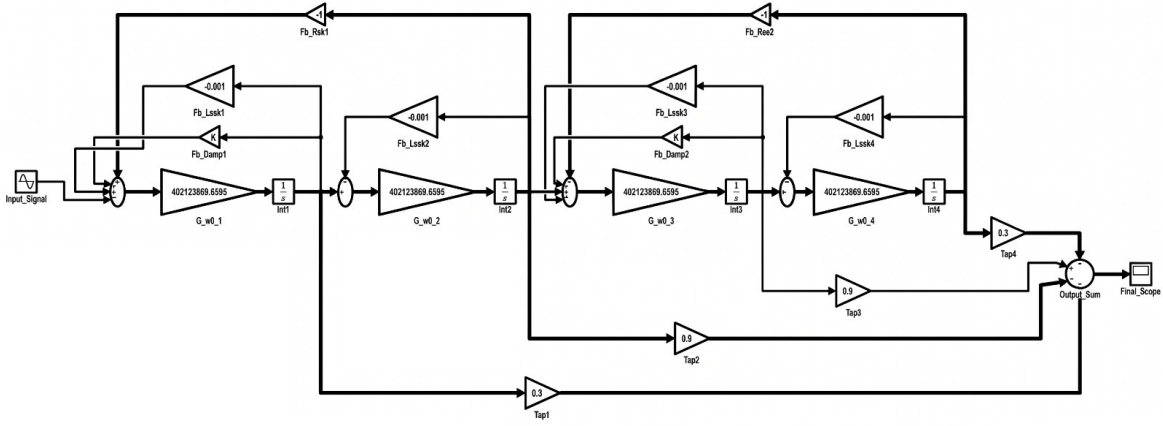


Fig. 2. Generated Simulink block diagram for a 4th-order CRFF filter, illustrating the cross-coupled resonator chains and feed-forward taps.

II. SOFTWARE ARCHITECTURE AND IMPLEMENTATION

The tool is developed using MATLAB's App Designer guides, utilizing an object-oriented approach. As shown in the active analysis view in Fig. 3, the interface is divided into an input parameter grid and a multi-tab visualization pane.

A. Analytical Engine and Parameter Linkage

At the core of the GUI is a dynamically linked parameter set consisting of Center Frequency (f_0), Target Quality Factor (Q), and Bandwidth. Modifying any single parameter recalculates the others to maintain consistency. Crucially, when the target Q or bandwidth is adjusted, the engine instantly calculates the minimum recommended DC gain $A_{vol}(dB) = 20 \log_{10}(20Q)$ to ensure stability and obtain a filter performance near theoretical values ($>90\%$ accuracy) [4]. Unless manually overridden by the user, this automatically derived gain is used for the non-ideality parameters in the transfer functions, Simulink models, as well as the Verilog-A model.

Using these variables, the analytical engine constructs the finite-gain State-Space matrix, executes a high resolution frequency sweep (50,000 points), and extracts the realized poles and zeros to populate the Pole-Zero map.

B. Simulink File Generation

One of the most error-prone aspects of system-level design is manually wiring complex feedback topologies. The tool mitigates this by generating a .slx model using `add_block` and `add_line` commands.

Beyond structural generation, the calculated parameters determine the block properties. The software converts the user's inputs into physical routing parameters: it calculates the ideal global damping factor ($-1/Q_{target}$) and the parasitic integrator leakage ($-1/A_{vol}$). These values are then injected into the respective feedback gain blocks within the Simulink model, ensuring that the time-domain schematic directly reflects the specified physical limitations.

C. Verilog-A Export

To bridge the gap between the filter model and a circuit-level design, the tool generates an `integratorA.va` file. This script dynamically writes a Verilog-A hardware description module, embedding the GUI's constraints directly into the code. Specifically, the required linear DC gain (A_{vol}) is written as a `dc_gain` parameter. Inside the Verilog-A `analog` block, this parameter is utilized along with the target Q to dynamically compute the total damping of the system ($\frac{1}{Q_{tun}} + \frac{1}{dc_gain}$). This export feature translates the synthesized specifications into a user friendly behavioral model, allowing IC designers to drop the block directly into Cadence Virtuoso for transistor-level testbenches.

Although it is quite straightforward to add summation blocks in Simulink, it is not as simple when dealing with voltage mode blocks. For this reason, the initial version of this tool exports current-mode integrators, giving the user the ability to directly connect the feedback and feed-forward taps in the Virtuoso environment.

III. THEORETICAL FRAMEWORK

A. Cascaded Resonator Feed-Forward (CRFF) Topology

The design tool utilizes the CRFF topology, which offers a significantly more forgiving design space than the CRFB architecture. The primary advantage of CRFF lies in its relaxed analog constraints; because the input signal is subtracted at the primary summation node, the internal resonators process predominantly quantization noise. Consequently, internal nodes experience minimal signal swing, allowing users to achieve stable initial system-level simulations without needing to immediately define the rigorous, highly linear OPAMP specifications that are mandatory in CRFB structures.

Furthermore, the CRFF architecture fundamentally simplifies the dynamic range scaling of state variables and reduces the feedback complexity. While CRFB designs require multiple feedback Digital-to-Analog Converters, introducing a multidimensional parameter space of timing mismatches and

intersymbol interference—a CRFF topology typically consolidates the feedback loop to a single DAC. This isolation of non-idealities, combined with straightforward coefficient scaling, abstracts away severe multi-path nonlinearities.

As a result, inexperienced designers can focus on validating their target Noise Transfer Function (NTF) and synthesizing the loop filter with a much higher probability of initial success before tackling transistor-level bottlenecks. [5].

The system is mathematically represented using a State-Space formulation:

$$\dot{x} = Ax + Bu, \quad y = Cx + Du \quad (1)$$

For a multi-order filter, the system matrix A is constructed as a chain of coupled feedback loops. The diagonal elements encapsulate the stage damping, while the off-diagonal elements represent the cross-coupling (energy transfer) between adjacent resonators. The C matrix dictates the feed-forward taps that create transmission zeros to shape the passband.

B. Modeling Integrator Non-Idealities

An ideal continuous-time integrator has the transfer function $H(s) = \omega_0/s$. However, physical implementations, such as the widely used Nauta OTA [6] in current-mode circuits, possess a finite DC gain (A_{vol}).

To predict real-world behavior, the tool substitutes the ideal integration step with a lossy integrator approximation:

$$H_{real}(s) = \frac{\omega_0}{s + \frac{\omega_0}{Q_{target}} + \frac{\omega_0}{A_{vol}}} \quad (2)$$

The added term $\lambda = \omega_0/A_{vol}$ represents the leakage caused by the finite amplifier gain, effectively acting as parasitic damping. By injecting λ term directly into the diagonal entries of the State-Space A matrix, the tool shifts the system poles exactly as they would shift in a physical circuit, allowing the designer to accurately observe and quantify this Q-reduction prior to SPICE simulation. Consequently, the realized quality factor ($Q_{realized}$) of a single resonant stage becomes a parallel combination of the intended damping and the amplifier's linear DC gain (A_{vol}):

$$\frac{1}{Q_{realized}} = \frac{1}{Q_{target}} + \frac{1}{A_{vol}} \quad (3)$$

Following standard design heuristics, setting the minimum DC gain to $A_{vol} = 20Q_{target}$ inherently introduces an approximate of 10% degradation in the Q-factor for a single isolated resonator. However, in multi-stage CRFF topologies, the phase errors induced by multiple lossy integrators accumulate across the cascade. The results extracted from the proposed tool demonstrate that for a 4th-order architecture, this compounding effect worsens the overall bandwidth widening that is even more apparent on 6th-order architecture. This was accounted for the built-in normalized transfer functions for 4th and 6th order systems, so the user will see a similar Q-reduction for all 3 orders filters having the same target Q, Bandwidth and DC Gain specifications.

C. Coefficient Selection and Q-Correction

When cascading multiple resonant stages to achieve higher-order filtering, the overall system bandwidth naturally shrinks. For instance, cascading two identical 2nd-order stages yields a narrower passband than a single stage.

The tool dynamically adjusts the internal stage quality factor (Q_{stage}). This modification ensures that the resulting system bandwidth strictly adheres to the user's initial specifications. By design choice, the target BW and Q parameters written in the input boxes of the GUI are the best theoretical values, meaning that only by selecting a higher Integrator Gain the actual filter performance will converge to these target values.

IV. SIMULATION RESULTS

To validate the tool, a 4th-order CRFF bandpass filter was designed. The target specifications are summarized in Table I.

TABLE I
TARGET SPECIFICATIONS FOR THE 4TH-ORDER FILTER

Parameter	Value
Center Frequency (f_0)	64 MHz
Target Quality Factor (Q)	50
Target Bandwidth	0.83 MHz
Minimum DC Gain (A_{vol})	60 dB

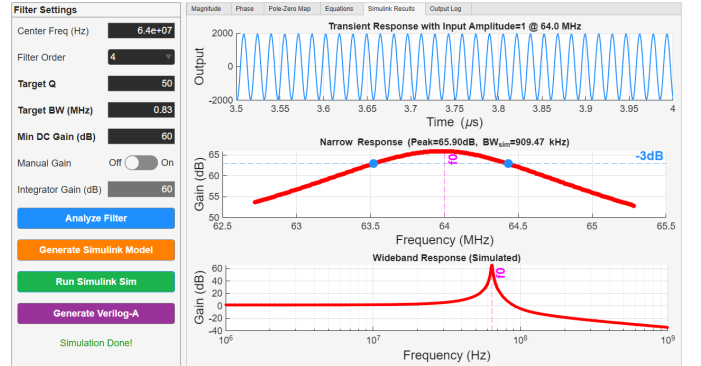


Fig. 4. Simulink transient and AC sweep results. The simulation runs a high resolution frequency sweep to display the corresponding Bode plot.

A. Analytical Verification

The generated magnitude response is shown in Fig. 3. By incorporating the 60 dB finite gain limitation, the analytical engine successfully predicts a slight Q-reduction. The -3 dB bandwidth is calculated as 905 kHz (due to finite gain), and the system confirms decent out-of-band rejection.

The stability of the filter can be verified in the Pole Zero Mapping tab in the GUI for both Analytical Filter and Realized Simulink Model. The finite gain leakage term pushes the poles slightly towards the left half-plane (LHP), increasing damping. The tool correctly identifies that all complex conjugate pole pairs remain safely within the LHP, ensuring a stable system.

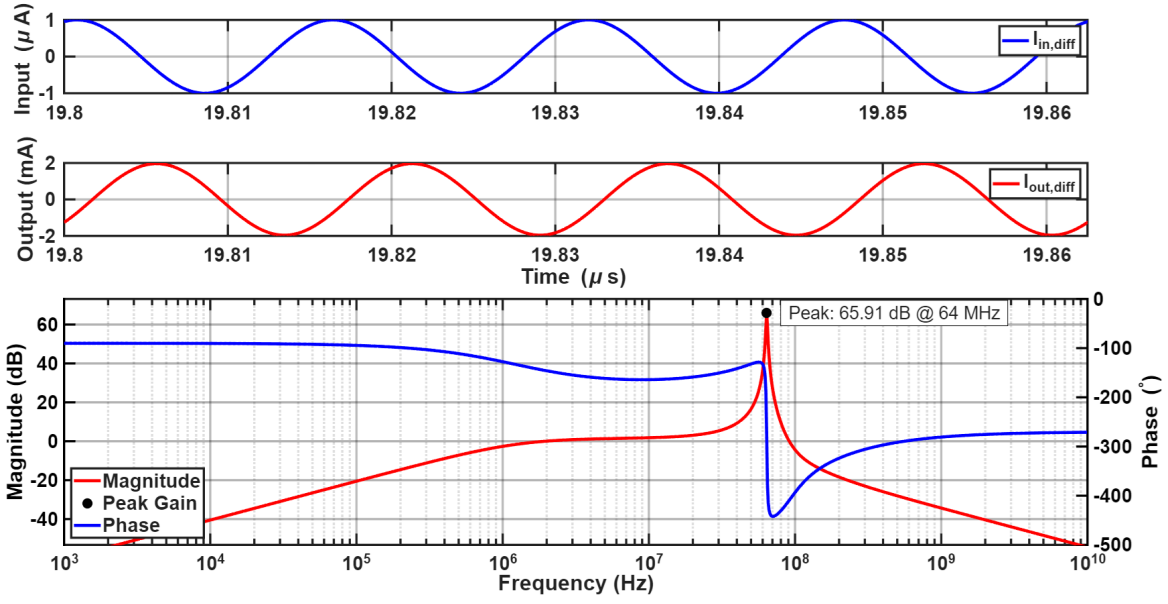


Fig. 5. Cadence Virtuoso AC and transient simulation results for the 4th-order filter model implemented using the generated Verilog-A behavioral models.

B. Transient and AC Simulink Verification

The Simulink model (shown in Fig. 2) was simulated using the Runge-Kutta fixed-step solver with a step size of 1 ns. The tool also runs a narrow passband sweep comprising 200 frequency points to capture the sharp resonance peak. The zoomed-in AC sweep (middle plot) accurately mirrors the Bode plot derived in the analysis stage, proving the reliability of both the State-Space formulation and the generated Simulink topology. All of these results are depicted in Fig. 4.

C. Cadence Virtuoso Verification via Verilog-A

To ensure a smooth transition from system-level modeling to circuit-level implementation, the Verilog-A code exported by the MATLAB tool was imported into Cadence Virtuoso, and a testbench was constructed by instantiating the Verilog-A blocks within the Virtuoso. AC and transient simulations were performed using identical configuration parameters (64 MHz center frequency, $Q = 50$, and 60 dB finite gain constraint).

The simulation results retrieved from Cadence matched both the MATLAB analytical plots and the Simulink transient data with a high degree of fidelity (as shown in Fig. 5). The matching frequency response (65.9 dB at 64 MHz) characteristics explicitly validated that the programmatic generation of the Verilog-A behavioral code accurately translates the modeled non-idealities into the EDA software ecosystem, ensuring that the designed topologies function exactly as predicted prior to the transistor-sizing phase.

V. CONCLUSION

This paper presented an automated design, modeling and analysis tool for high-order continuous-time bandpass filters. By actively incorporating integrator non-idealities into the State-Space mathematical synthesis, the tool allows designers to accurately predict bandwidth widening and Q-factor

degradation before beginning a circuit design and layout. Furthermore, generated Simulink blocks and Verilog-A models significantly reduces manual drafting errors and interlinks the system-level architecture with the physical IC design.

While the current iteration of the tool provides a comprehensive workflow, there is still room for improvement. Future development will focus on integrating custom normalized transfer functions, enhancing the robustness of the automated block diagram placement algorithms, and adding support for Cascaded Resonator Feedback (CRFB) topologies alongside the existing CRFF structures. Additionally, voltage-mode integrators, finite slew rate, custom input and output resistances options are currently being explored. All of these updates are aimed to be implemented in future releases and will be presented as an open-source tool for all interested parties in the IC design community.

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